

Math Camp 2025 – Problem Set 3

Read the following problems carefully and justify everything you do.

1. Limits. Find the following limits or show that they don't exist.

1. $\lim_{x \rightarrow +\infty} \frac{e}{x}$.

2. $\lim_{x \rightarrow -\infty} \frac{e}{x}$.

3. $\lim_{x \rightarrow 3} \frac{x}{x^3 - 27}$.

4. $\lim_{x \rightarrow 3} \frac{x - 3}{x^3 - 27}$.

5. $\lim_{x \rightarrow \infty} \frac{x + 1}{2x}$.

6. $\lim_{x \rightarrow \infty} \left(\frac{1}{2}\right)^x$.

7. $\lim_{x \rightarrow \infty} \frac{3x^3 + 2x^2 - x + 3}{4x^4 + 3x^3 + 2x^2 + x + 4}$.

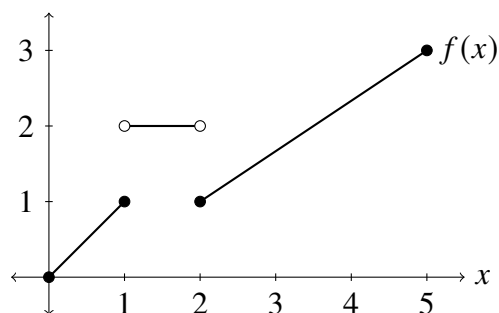
8. $\lim_{x \rightarrow 0} \frac{1}{x^2}$.

9. (a) $\lim_{x \rightarrow 1} f(x)$.

(b) $\lim_{x \rightarrow 2} f(x)$.

(c) $\lim_{x \rightarrow 5} f(x)$.

where $f : [0, 5] \rightarrow \mathbb{R}$ is given by the following graph.



2. Continuity.

1. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ x & \text{if } x \geq 1. \end{cases}$$

Is it continuous?

2. The following functions are defined for all $x \in \mathbb{R}$ except for one point x_0 . Find x_0 and determine if they can be defined at x_0 so that they are continuous on $\mathbb{R}$.

(a) $f(x) = \frac{x-3}{x^3-27}$.

(b) $f(x) = \frac{1}{x}$.

(c) $f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ x & \text{if } x > 1. \end{cases}$.

(d) $f(x) = \begin{cases} x^2 & \text{if } x < -1 \\ x & \text{if } x > -1. \end{cases}$.